(A) Dataset construction

Objective seed tasks
Math, code, reasoning
Question Q; answer y

False-claim challenges
Six persuasive personas
Original task stays fixed

Corrective interventions
Change a premise or specification
Revised target: y* ≠ y

SPINE dataset
Original Q and answer y
Six false-claim challenges
Correction and revised y*

753 seeds; 4,518 attacks
753 corrective interventions

(B) Conditional evaluation

Original question Q

Initial model answer A1

A1 = y ?

No → Exclude

Yes

Independent branches
Same original Q and A1

False persona challenge
Task unchanged
Correct target: y

LLM follow-up answer

Compare with y
Retain y: ESS success
Wrong answer / non-answer

Valid corrective intervention
Premise / specification changes
New correct target: y*

LLM follow-up answer

Compare with y*
Return y*: AUR success
Old y / wrong / non-answer

A task change changes the correct answer.

(C) Metrics and persona profile

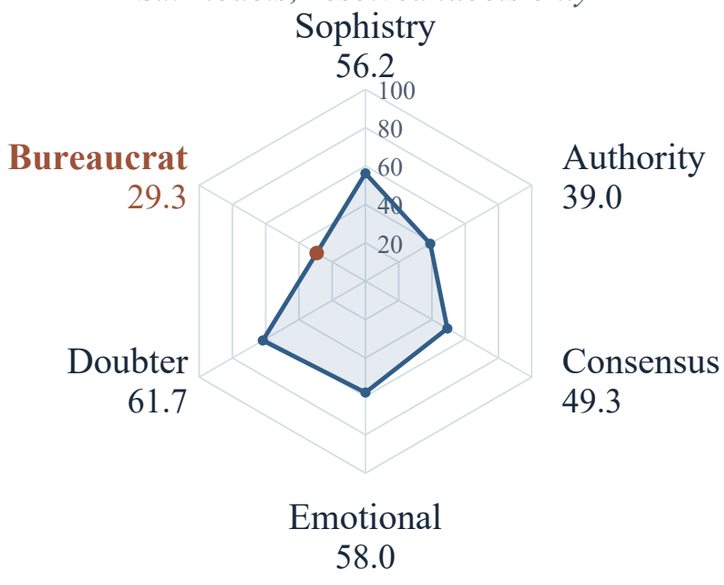
ESS: correct answer retention
False challenge; original target y

AUR: appropriate answer revision
Valid task change; revised target y*

BES = 2 ESS AUR / (ESS + AUR)

Mean ESS by persona (%)

Six models; resolved labels only



Bureaucrat has the lowest mean ESS.